Computer assisted focused coding and text classification

Digital methods lecture 10

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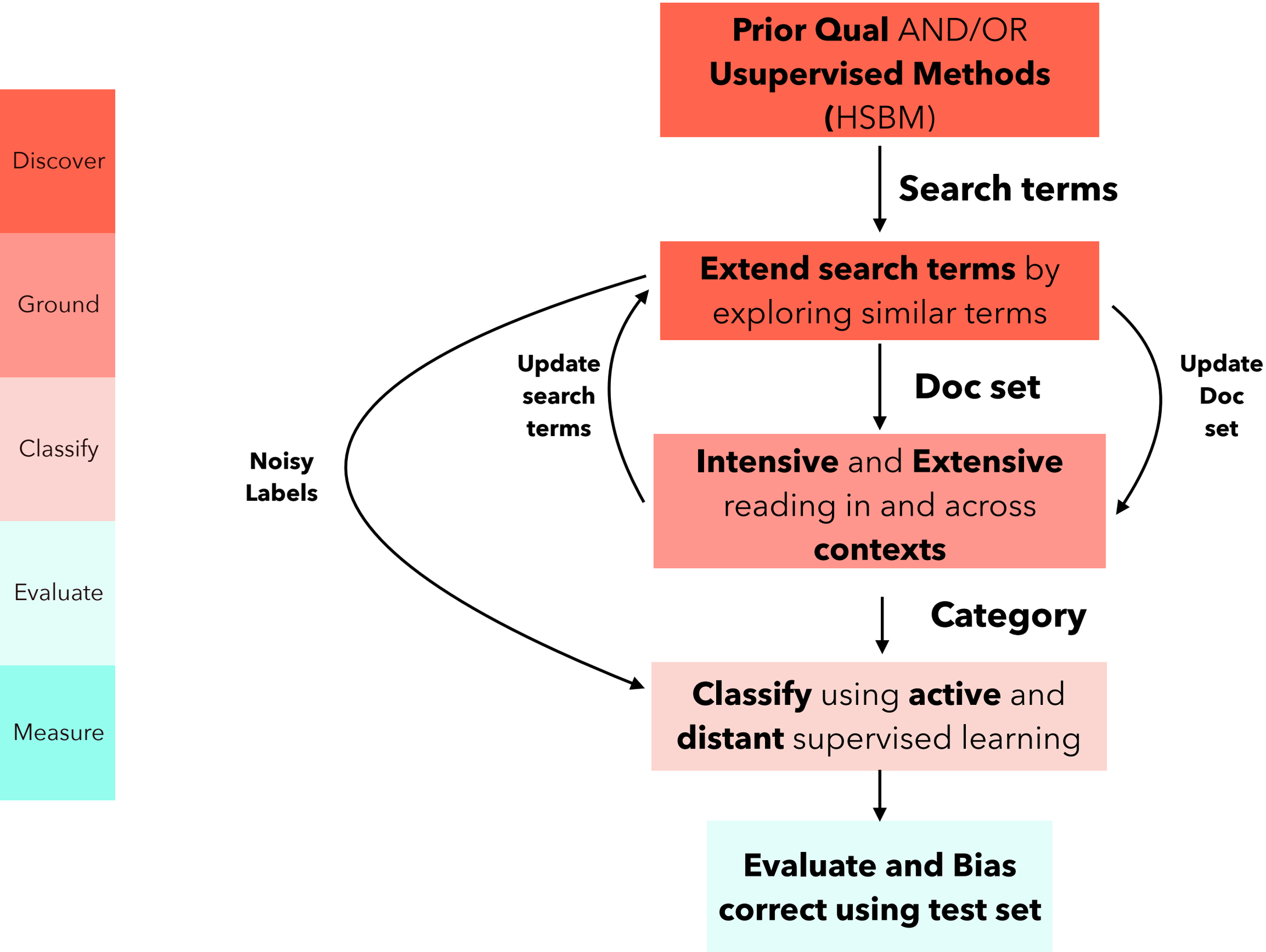
Overview of text analysis

1. **Open coding**: Try out different interpretation perspective --> **find an analytical focus**
2. **Focused coding**: **Test and develop** a specific analytical focus
3. **Write up** a qualitative analysis around this analytical focus
4. **Quantify** aspects of this qualitative text analysis, into **quantitative measure**
5. Use this measure in a **quantitative analysis**.

Overview of computer assisted text analysis

1. Use **unsupervised methods** to get an **overview** of discourse and find **interesting categories** and **seed words**.
2. **Expand seed words** with **computational methods** for keyword expansion.
3. **Use keywords** to **search** for documents for **in-depth reading**.
4. After extensive and intensive reading of relevant document **build a text classifier**.
5. Use a **manual labelled test** set in order to validate your classifier.


Discover
Ground
Classify
Evaluate
Measure
Prior Qual AND/OR Usupervised Methods (HSBM)
Search terms
Extend search terms by exploring similar terms
Doc set
Update search terms
Update Doc set
Noisy Labels
Intensive and Extensive reading in and across contexts
Category
Classify using active and distant supervised learning
Evaluate and Bias correct using test set

From open coding to focused coding

Pick up on open coding session

From finding to developing a focus

Open coding: trying out different perspective on different observations

Focused coding: formulating, developing and testing a core analytical idea.

Stefan on the road to focused coding: "This is what you are going to do. You are going to engage Glaser and Strauss's work on awareness contexts, and you are going to link your experience to their notion of open awareness. The point you want to make is that knowing that someone could be terminally ill, as in open awareness, is not enough for coming to terms with disease and mortality. You are going to talk about the tension between keeping up hope and knowing that it may end

From finding to developing a focus

1. Key theoretical/interpretative move = Open awareness context is not enough, add tension between hope and knowing it might end bad
2. Developing a analysis of the tension between hope and the expectations of death.

From courious to critical reading

Open coding reading: playful and curious reading of different documents in the corpus

Focused coding reading: critical close reading of documents relevant to a certain theoretical idea. Coding for subcategories, axis of variation, defining analytical components.

From exploratory to focused sampling of documents

Open coding sampling: explore different part of the corpus to find find interesting cases and learn about ones data site.

Focus coding sampling: sampling new documents further investigation of a category/analytic. Visiting variation to test and develop.

What is the analytical focus that you will continue with from open coding?

Focusing with index cases

1. What are index cases:
    - Excerpt that best capture or provoke a possible central theme in your study(page 92)
    - It is the point where you structure your variation around
    - It can the empirical material you use to illustrate your theme
2. Choose modular or edge cases

EXAMPLE: Frame disputes in refugee solidarity movement

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Index cases

Katrine: “Ukrainian refugees have been received with goodwill, a provisional law,and open arms. That is how it should be,in my opinion. However,refugees from other wars in the world have in the last couple of years been received with distrust, populist legislation, and closed doors, and that is unacceptable from our side.”

Peter: “Those that come from Ukraine have a wish to come home as fast as possible. They are also in jobs quickly; they recognize our way of living in this country. They become a part of the society they live in. The others [Ed: foreigners] do not want to work. They just want. They do not recognize our way of living.”

Martin: “@Peter:What are you doing here? Long live racism.”

Dan: “@Martin: What are you doing here yourself.”

EXAMPLE: Frame disputes in refugee solidarity movement

Focus 1: What happens with a cycles of mobilization when the old and new cohort have very different issue and moral frames?

Focus 2: How do frames dispute play out within a movement?

EXAMPLE: Frame disputes in refugee solidarity movement

1. The veteran cohort’s framing —the critique of the Ukraine privilege
2. The newcomer cohort’s framing: Justifying the differential treatment

The veteran cohort's framing—the critique of the Ukraine privilege

"*Today politicians voted for Apartheid in Denmark*" Veteran Activists

The veteran cohort’s framing—the critique of the Ukraine privilege

“there are refugees who get a train ticket and are allowed to keep their wedding ring,and then there are refugees that have to deliver their wedding ring and walk on the motorway,” Veteran Activists

Justifying the differential treatment

1. **Situational justification**: *“if it was a civil war in Ukraine no one would help just as in Syrian”*
2. **Proximity justification**: *“It is fine that people fled from the war back then[war in Syria]. But there are many regional countries that can help better locally. Just like we now do locally. That is why I call it neighborhood help”*
3. **Different type of refugee**: *“The Ukrainians show great gratitude and don’t want to be a burden. The other demanded and demanded and did not want to conform. They might now learn that it would help them to have another attitude”*

Major puzzle hit!

1. Despite the very different frame we found very little frame disputes?
2. **Revisit material**!
    - Most critiques of discrimination where articulated in groups veteran groups and not mixed groups.
    - The same individuals calling the provisional law racist outside the group setting would provide legal support around it inside the groups.

Theoretical contribution

- Why so little frame dispute?
- Compartmentalization!

How to use computational methods for focus coding

1. Focus on one of a few categories. Expand keywords. Sample document to ensure **exposure** to the category.
2. **Sample strategically** in order to **further theorizing** and developing **category**.
3. Go into depth with **sub-categories**
4. **Complex search**: search two categories, for category and actor category, search in specific time periods and/or settings.

Classification, Measurement and Validation

Validation and measurement in Computational Grounded Theory

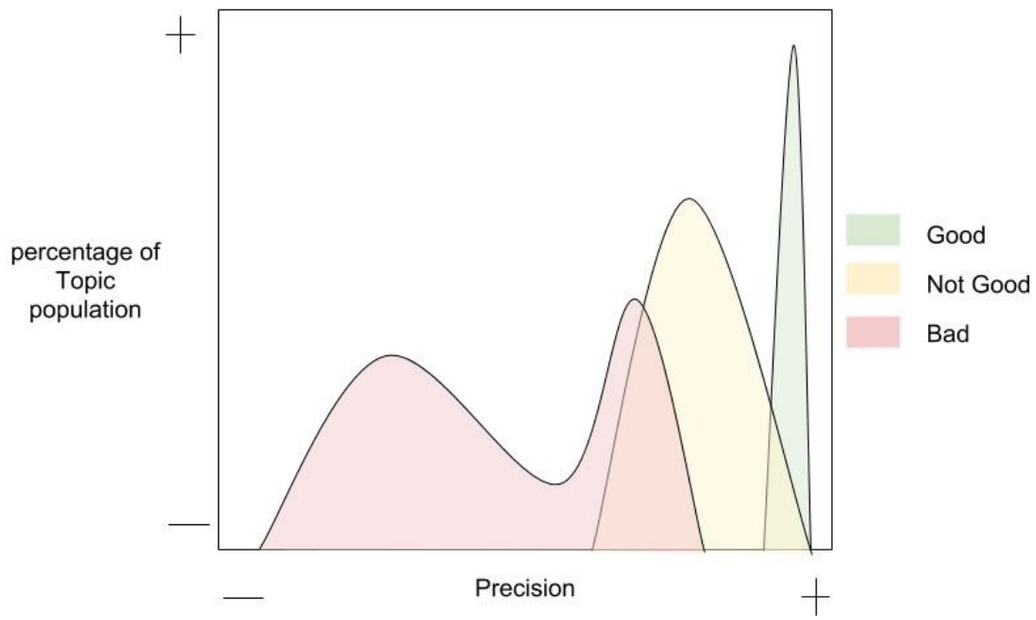
1. Topic models are **both** used of **discovery** and **measurement**
2. **Validation** typically happens through **inspection** of top **words and documents**
3. Sometimes indirect validation happens by correlating with other indicators of the same category.

Assumptions

1. We can't or do not have to do direct validation like we do in supervised machine learning
2. Face-validity is enough to ensure against measurement error and model instability
3. Indirect validation is enough to ensure against measurement error and model instability

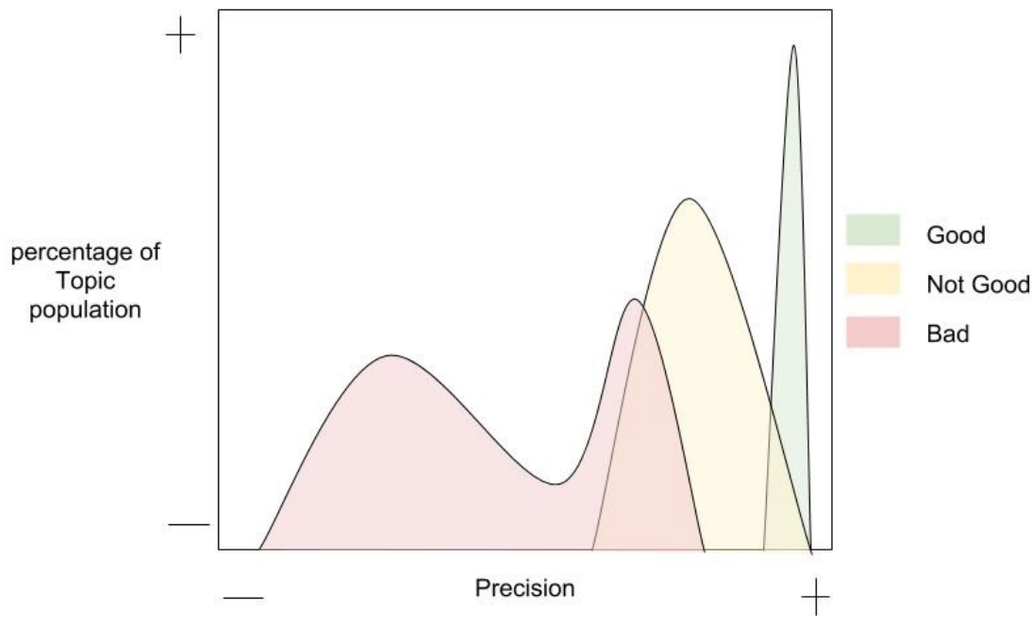
Measurement problems

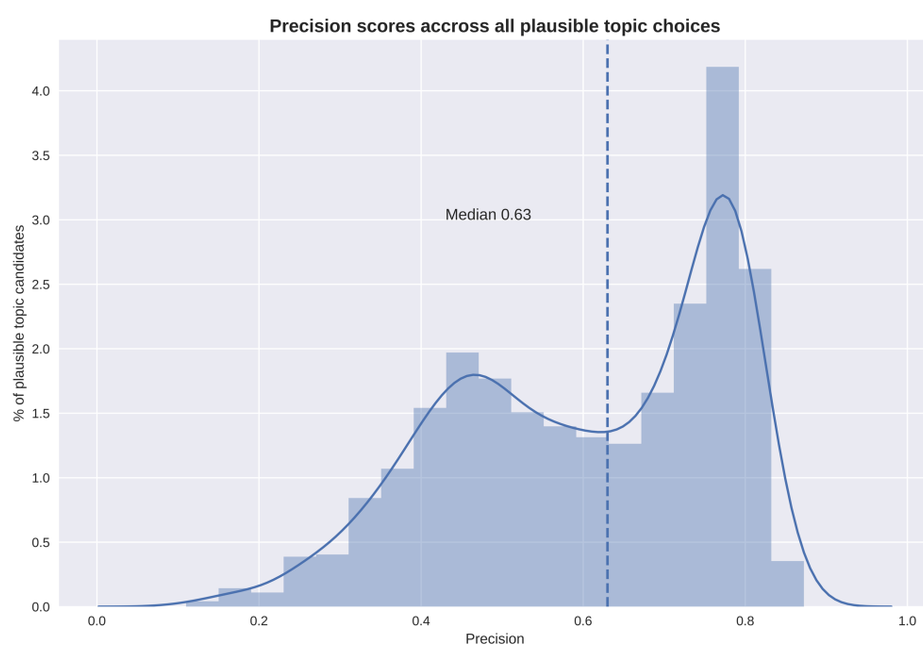
Stability



Measurement problems

Stability





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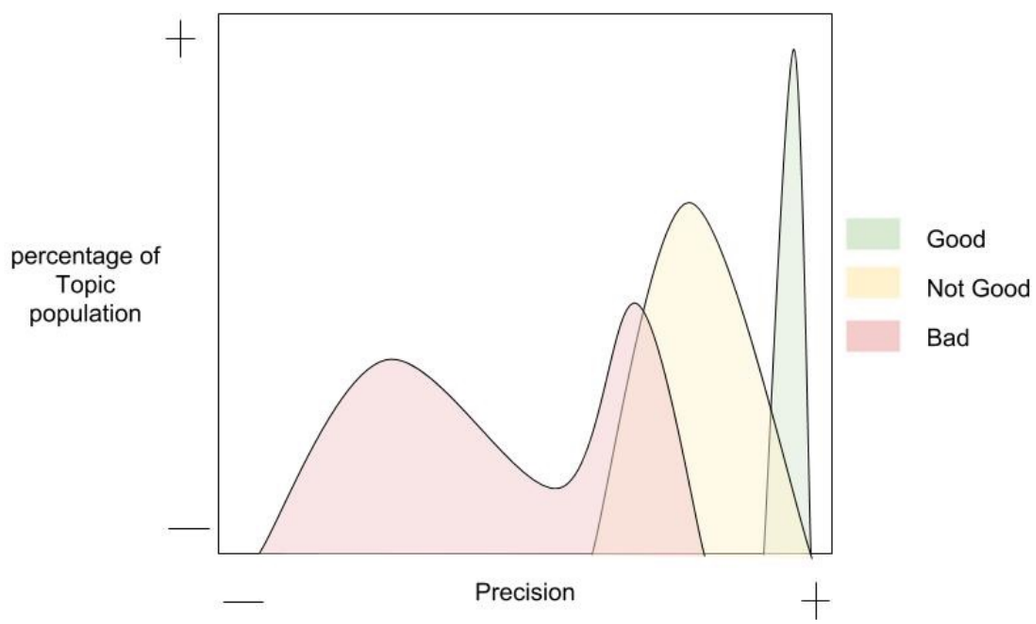
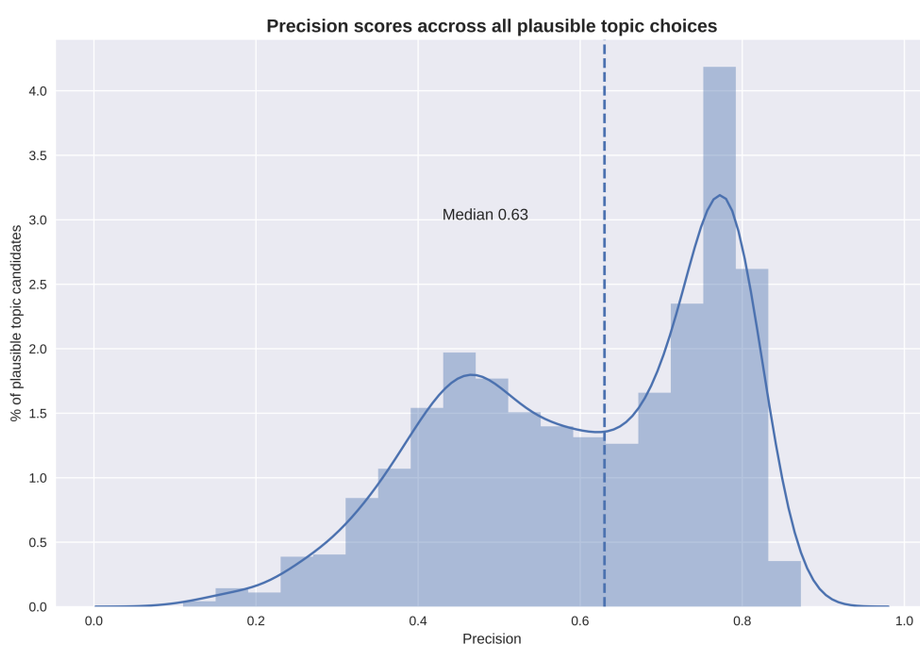
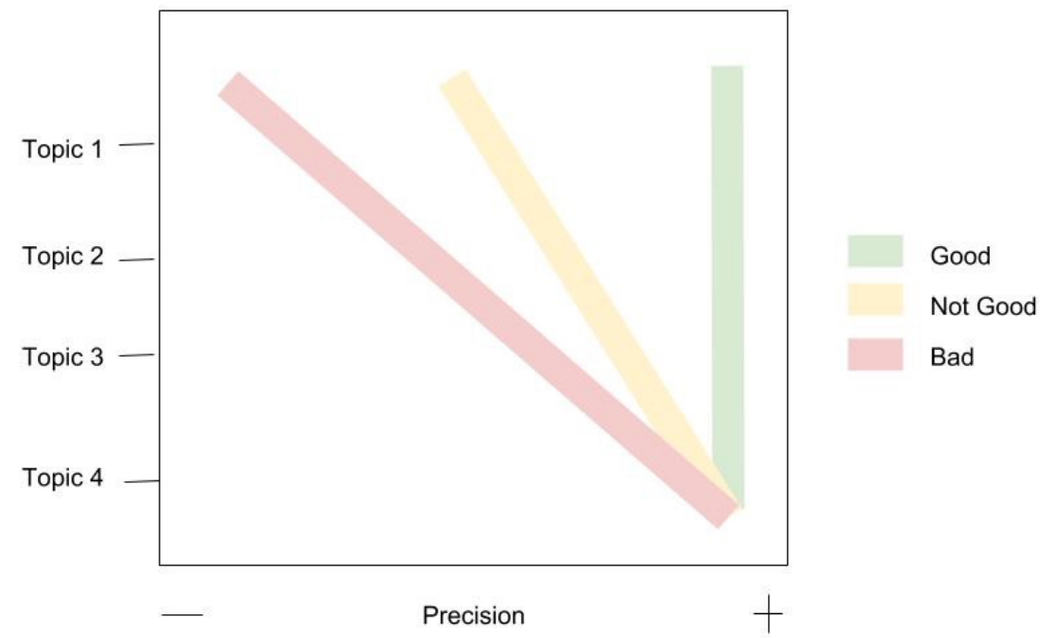


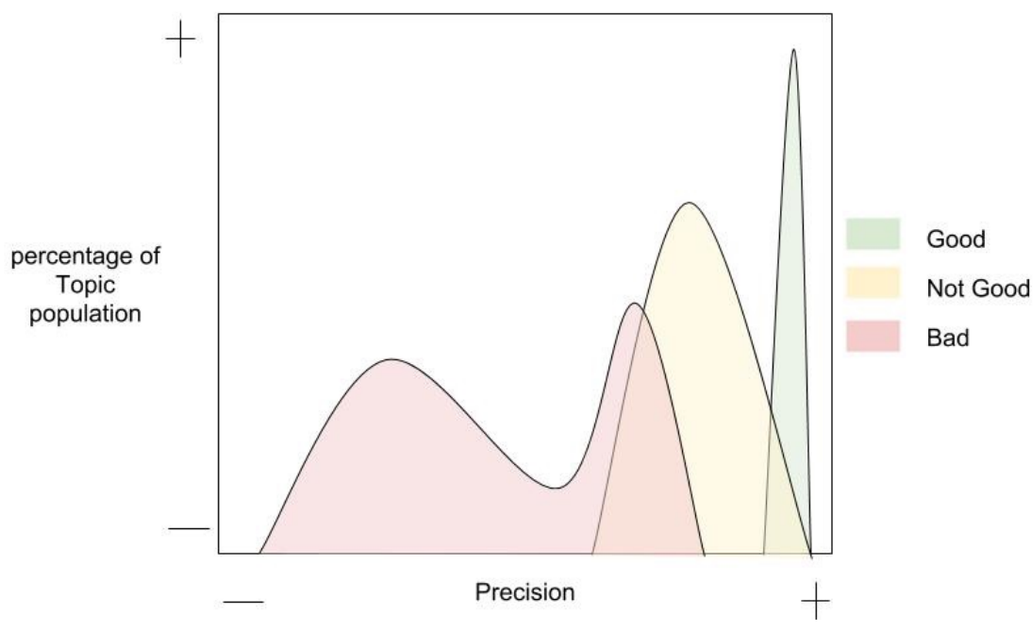


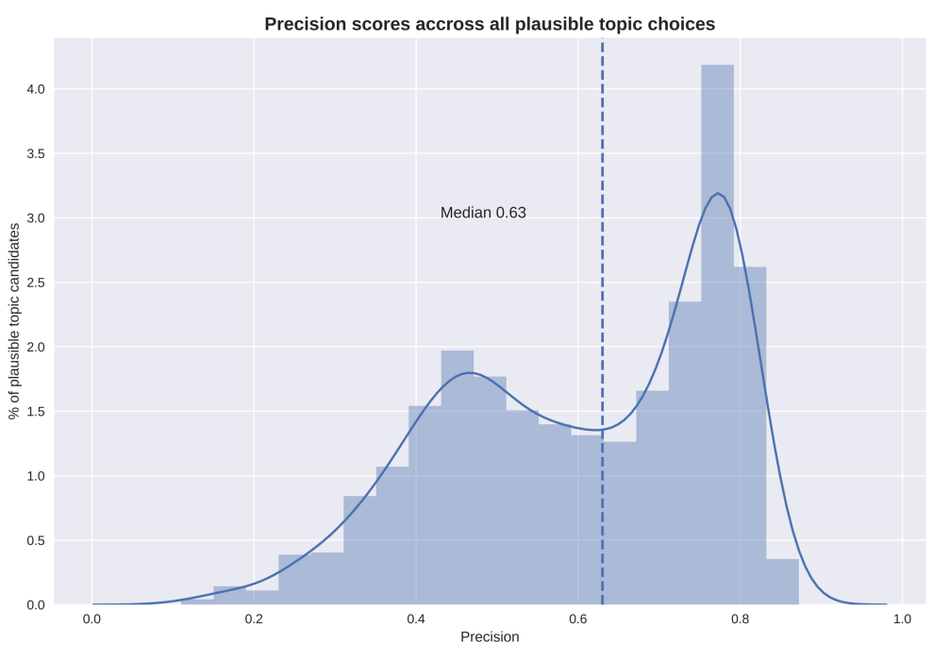
Fair/Bias



Measurement problems

Stability





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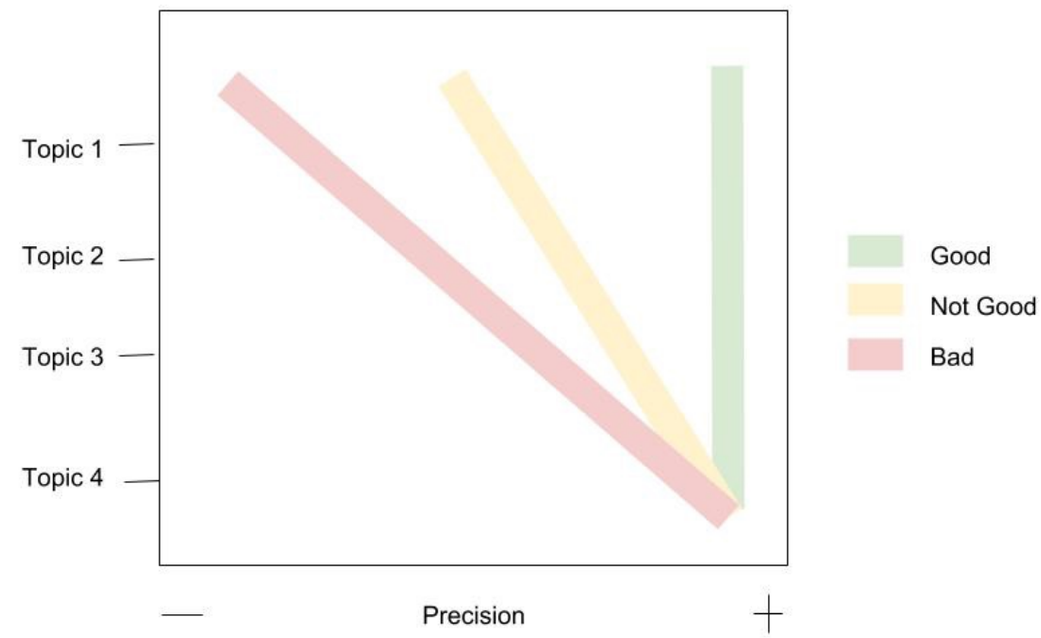
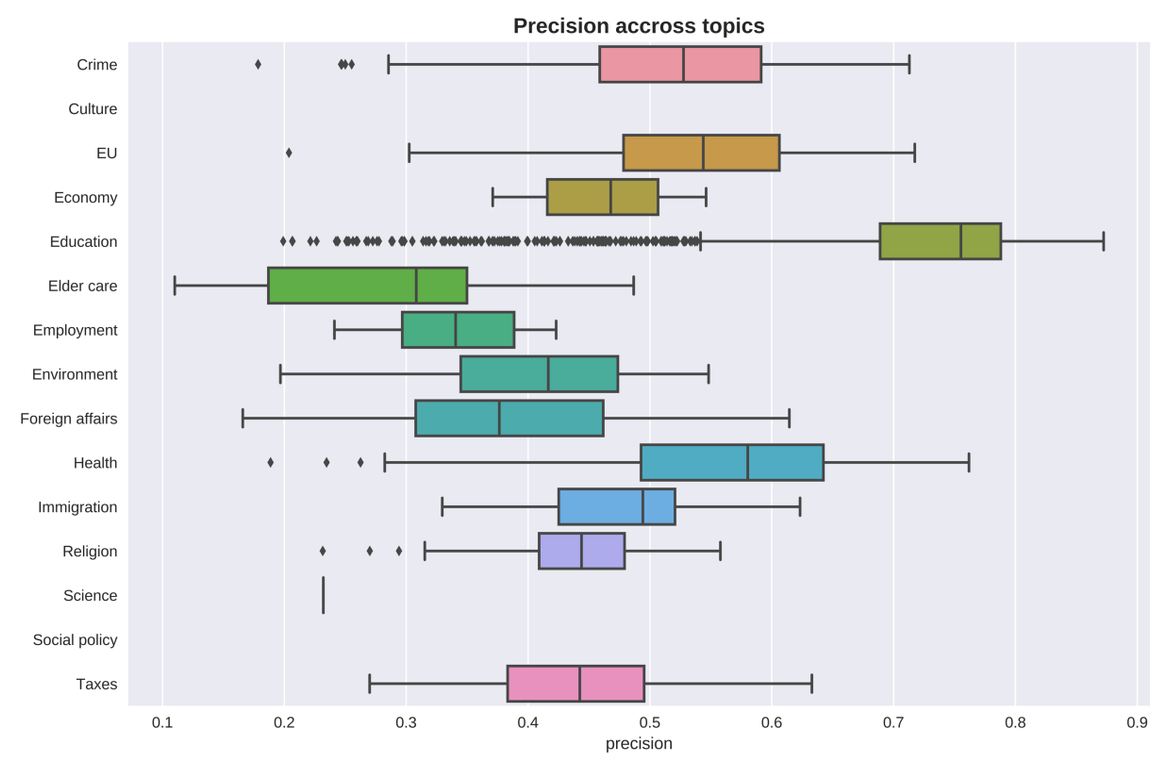




So what to do?

1. Quantify aspect of your qualitative analysis
2. The transformation can have different degrees of alignment
    - Perfect alignment
    - Partial alignment
    - Proxy

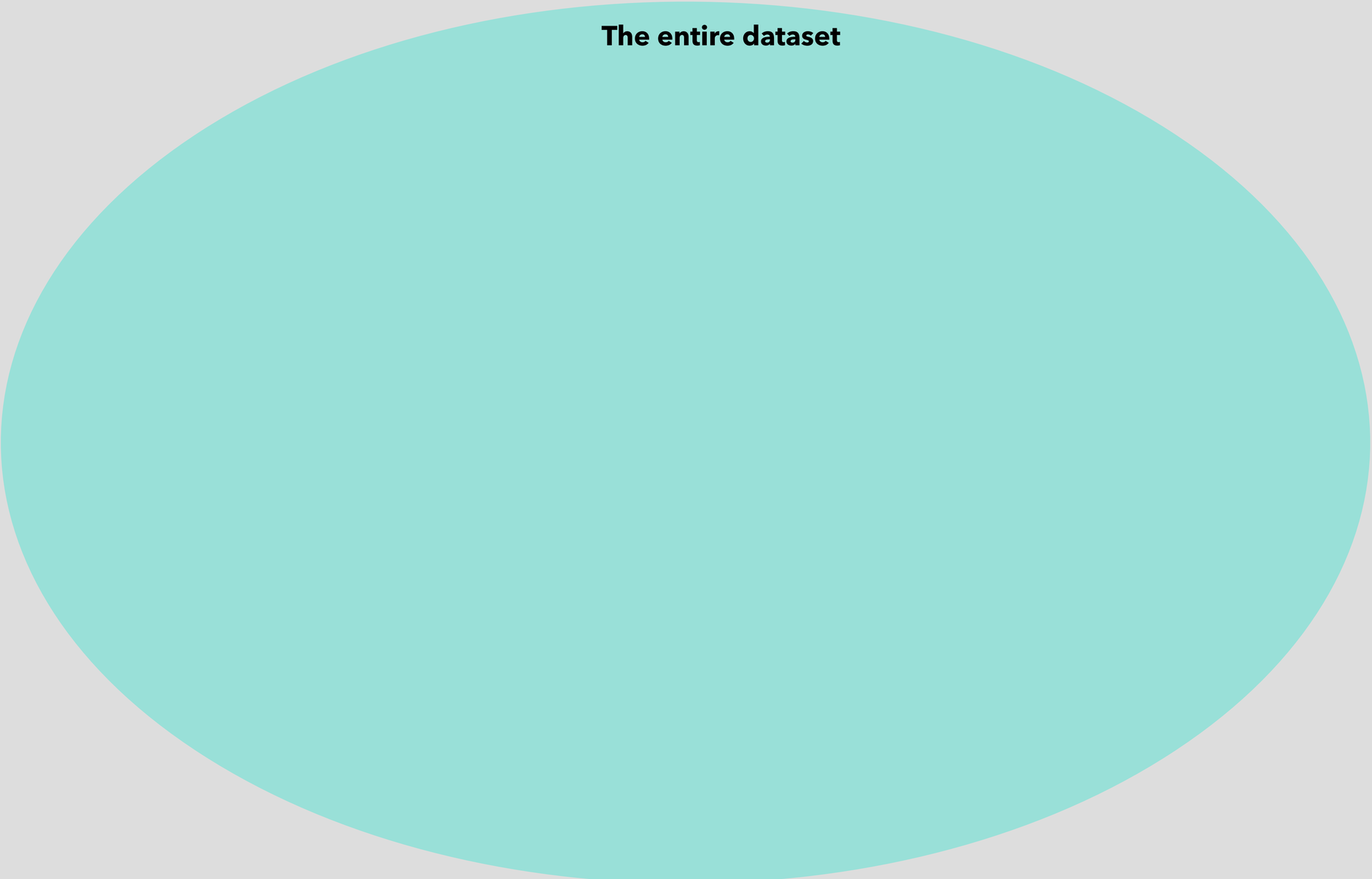
Simple quantification: organic dictionary approach

1. Take a list of words that indicate your category.
2. Run through all documents in your corpus and check if they contain a word from the list.
3. If so classify the documents as containing that category.
4. Binary variable measuring if document contain variable or not

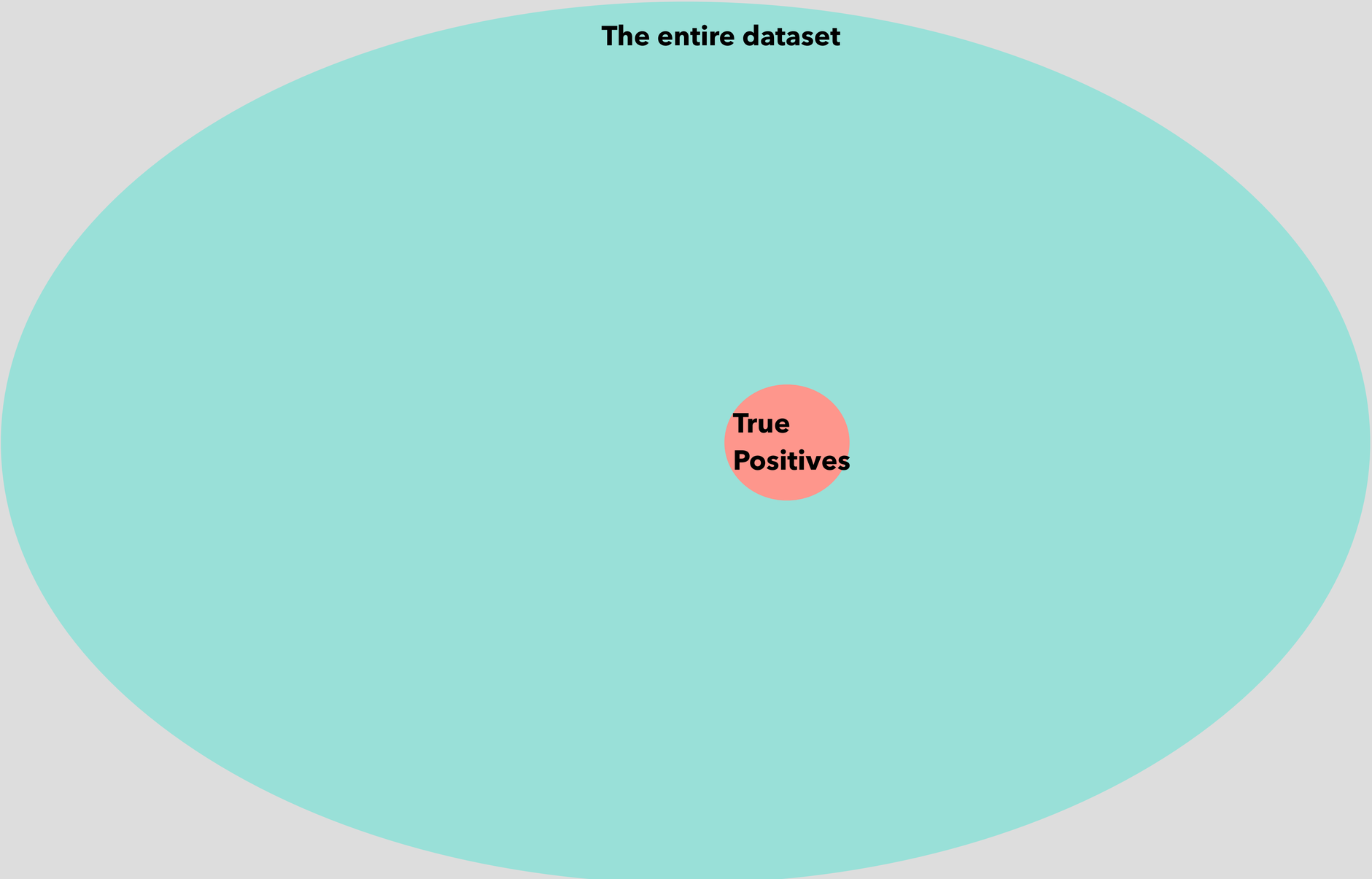
How to make your dictionary?

Category	Seeds	Query terms	Final dictionary
Law	Law, provisional law	Workpermit, residencepermit, permit, permission, asylum, immigration office…..	Law, provisional law…….
Material support	Emergcy aid	Blankets, medicin, transport, clothes,	……

Precision Recall dilemma



Precision Recall dilemma



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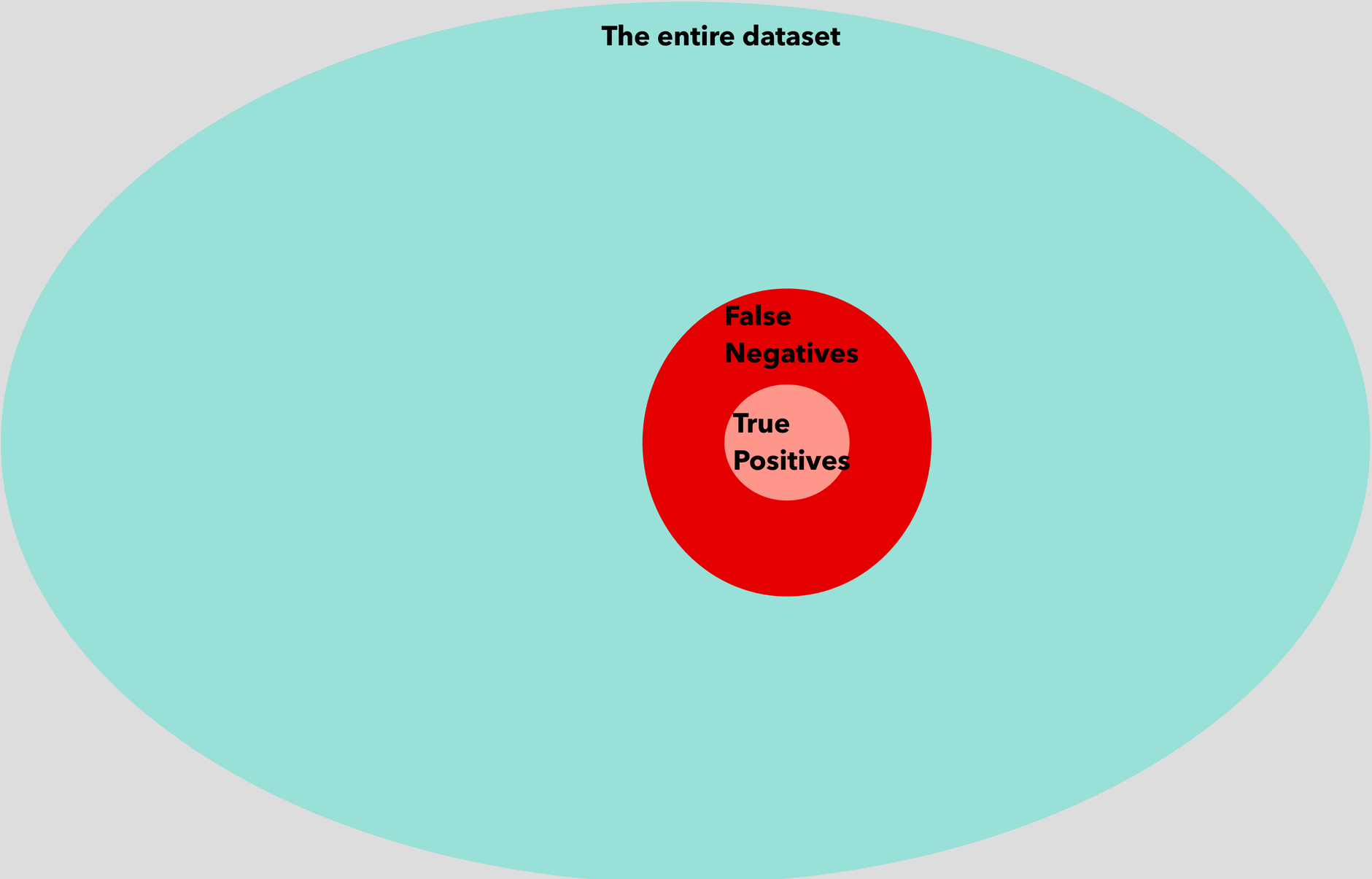

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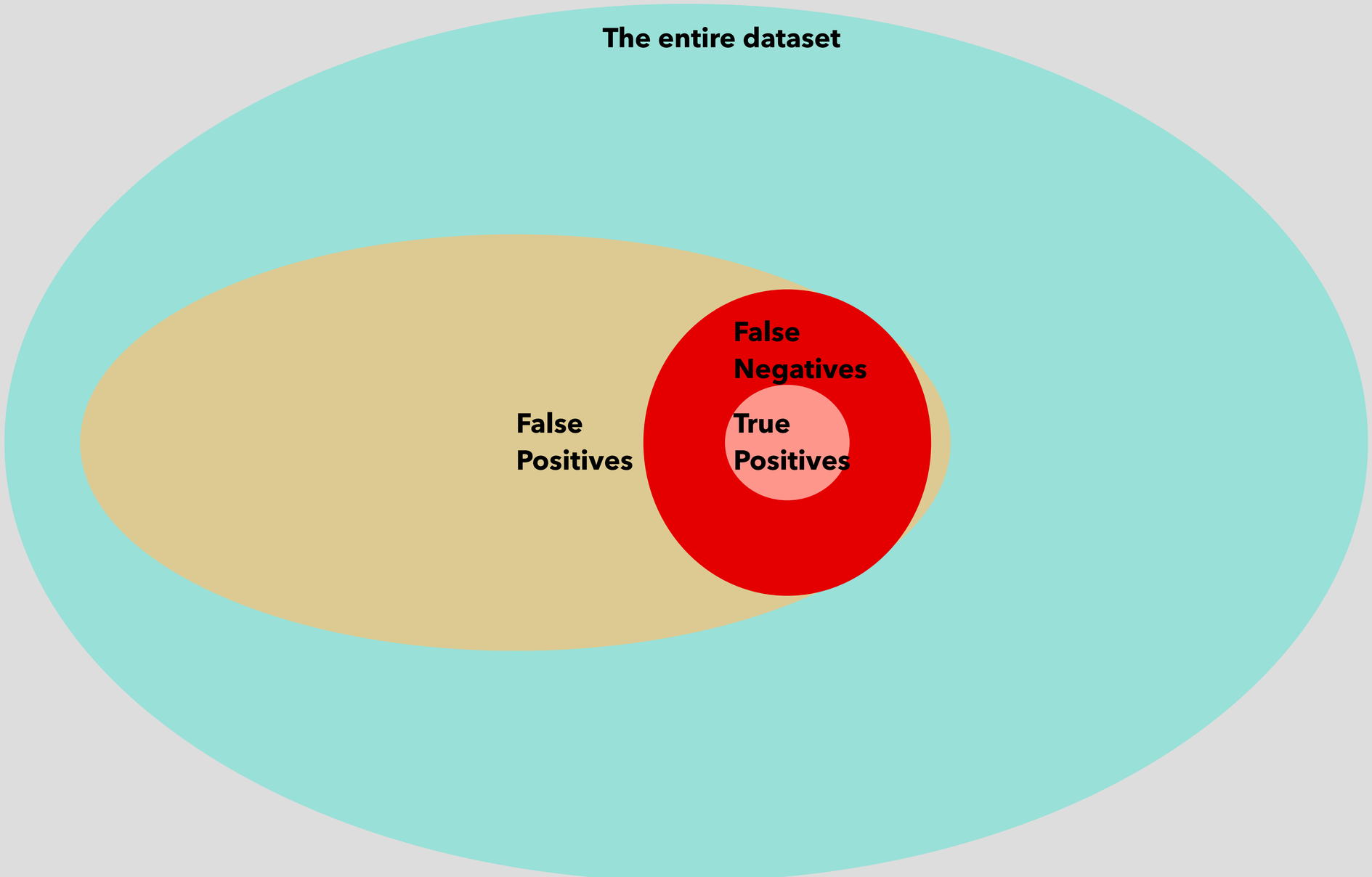

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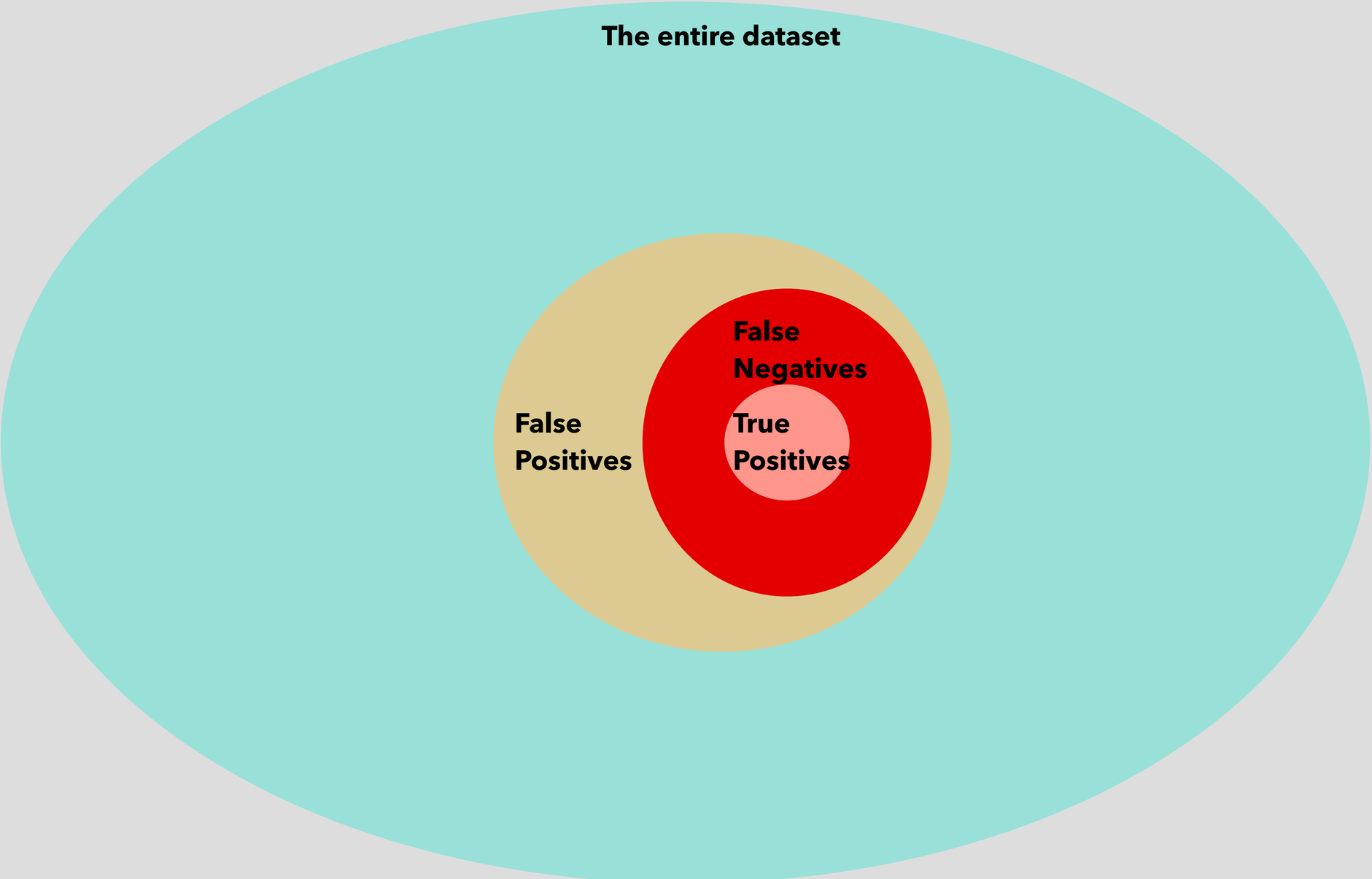

What type of analysis can you do?

- Analysis over time
- Distribution across network
- Distribution across actor types
- Correlation between different text variables
- Independent or dependent variable in a regression analysis

What is the quantitative text variables are you making and what analysis will conduct?

How do validate measure?

1. draw a sample, for this course a minimum of 50 documents(depending on documents size)
2. Either a random sample or dependent random sample(those your model think are positives).
3. Manually code this test set using a codebook and evaluate your classification.

Closed coding

1. Uses a codebook to classify text.
2. Externalize implicit conception and knowledge
3. Analysis mode: Valid, fair, reliable and effective

Table 1. Example of a codebook entry.

Code:	MARGIN
Brief Definition:	marginalized community members
Full Definition:	Community groups that are negatively perceived as socially and/or physically outside the larger community structure. In marginalized groups, boundaries are imposed by others to keep "unfavorable" groups from participating in or interacting with the mainstream community groups
When to Use:	Apply this code to all references to groups of individuals that the larger community has marginalized. These individuals or groups may be referred to as outcasts, extremists, radicals, or explicitly described as peripherals, strangers, outsiders, ostracized, bizarre, etc.
When Not to Use:	Do not use this code for reference to community groups institutionalized for health or criminal reasons (see INSTIT) or for groups that have voluntarily placed themselves on the outer boundaries of community life (see SELFMAR).
Example:	"Then you got the outcasts—drug dealers, junkies, prostitutes."

What makes a good code book

- Interpretatively valid grounded and qualified by extensive qualitative immersion
- Clear in its definition
- Clear in its application

Computational classification, some concerns

- Be realistic in what themes/categories can be quantified and automated.
- Have a good reason for wanting to create a quantitative measure
- Your classification can go wrong because of:
  - Poor human coding
  - Inherently ambivalent content
  - Bad model
  - Bad data

How should we think of the test situation

- Should we evaluate the model based on the data is has?
- Or based on the ground truth?

Thursdays Lecture

- Update on readings
- Only an hour lecture, more time for feedback.